
Linear Statistical Models

B. Statistical Data Science 2nd Year Indian Statistical Institute

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Exercise Series 3

Exercise 1 (Graded). We have seen the implications of the distributional assumption $\epsilon \sim \mathbf{N}_n(\mathbf{0}, \sigma^2 \mathbf{I})$. Under this assumption, $\hat{\boldsymbol{\theta}}_{\text{LSE}} \sim \mathbf{N}_d(\boldsymbol{\theta}, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1})$. Under our modeling assumptions, i.e., $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \epsilon$ with $\mathbf{X}$ non-random, the distributional assumption (i.e., $\epsilon \sim \mathbf{N}_n(\mathbf{0}, \sigma^2 \mathbf{I})$) also implies that $\mathbf{y} \sim \mathbf{N}_n(\mathbf{X}\boldsymbol{\theta}, \sigma^2 \mathbf{I})$. Hence we may try to find the maximum likelihood estimator (MLE) of $\boldsymbol{\theta}$ under this distributional assumption.

- (a) Write the likelihood and log-likelihood of the parameters $\boldsymbol{\theta}$ and σ^2 under the setup described above.
- (b) Find the MLE of $\boldsymbol{\theta}$ and σ^2 from part (a).
- (c) Is there any connection between the MLE and the LSE of $\boldsymbol{\theta}$? What about the estimators of σ^2 ?

Note: We have seen that $S^2 := \frac{\|\hat{\mathbf{e}}\|^2}{n-d}$ is an unbiased estimator of σ^2 , where $\hat{\mathbf{e}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\theta}}_{\text{LSE}}$ is the error of the model. $\|\hat{\mathbf{e}}\|^2 = \sum_{i=1}^n \hat{e}_i^2$ is the sum of squared errors or SSE.

- (d) Calculate the expectation and variance of the MLEs.

Exercise 2. In the class, we have seen that if $\mathbf{Z} \sim \mathbf{N}_n(\mathbf{0}, \mathbf{I})$ and $\mathbf{P}_{n \times n}$ is an orthogonal projection matrix (i.e., $\mathbf{P}^\top = \mathbf{P}$ and $\mathbf{P}^2 = \mathbf{P}$), then $\mathbf{Z}^\top \mathbf{P} \mathbf{Z} \sim \chi_r^2$, where $r = \text{rank}(\mathbf{P}) = \text{trace}(\mathbf{P})$. The essential step in that proof was the spectral decomposition of a projection matrix: $\mathbf{P} = \sum_{i=1}^r \mathbf{u}_i \mathbf{u}_i^\top$, where $\mathbf{u}_1, \dots, \mathbf{u}_r$ are orthonormal vectors, so that $\mathbf{u}_1^\top \mathbf{Z}, \dots, \mathbf{u}_r^\top \mathbf{Z} \stackrel{\text{i.i.d.}}{\sim} \mathbf{N}(0, 1)$. Here, we generalize this result for general p.s.d. matrices.

- (a) Let $\mathbf{A}_{n \times n}$ be a p.s.d. matrix. Redo the steps and verify that $\mathbf{Z}^\top \mathbf{A} \mathbf{Z} = \sum_{i=1}^n \lambda_i W_i^2$, where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of $\mathbf{A}$ and $W_1, \dots, W_n$ are i.i.d. $\mathbf{N}(0, 1)$ random variables. So, for any p.s.d. matrix $\mathbf{A}$, $\mathbf{Z}^\top \mathbf{A} \mathbf{Z}$ is a weighted sum of independent χ_1^2 random variables.

Hint: Use the spectral decomposition of a p.s.d. matrix: $\mathbf{A} = \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}^\top = \sum_{i=1}^n \lambda_i \mathbf{u}_i \mathbf{u}_i^\top$, where $\mathbf{U}$ is an orthogonal matrix (i.e., $\mathbf{u}_1, \dots, \mathbf{u}_n$ are orthonormal) and $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$.

- (b) Show that $\mathbb{E}(\mathbf{Z}^\top \mathbf{A} \mathbf{Z}) = \text{trace}(\mathbf{A})$ and $\text{Var}(\mathbf{Z}^\top \mathbf{A} \mathbf{Z}) = 2 \text{trace}(\mathbf{A}^2)$.

Hint: Use part (a). What are the expectation and variance of a χ_1^2 random variable?

We now extend this result to normal random vectors with general covariance structure.

- (c) Let $\mathbf{Z} \sim \mathbf{N}_n(\mathbf{0}, \Sigma)$ where $\Sigma \succeq \mathbf{O}$ and $\mathbf{A}_{n \times n}$ is a p.s.d. matrix. Show that $\mathbb{E}(\mathbf{Z}^\top \mathbf{A} \mathbf{Z}) = \text{trace}(\mathbf{A}\Sigma)$ and $\text{Var}(\mathbf{Z}^\top \mathbf{A} \mathbf{Z}) = 2 \text{trace}((\mathbf{A}\Sigma)^2)$.

Hint: $\mathbf{Z} \stackrel{\mathcal{D}}{=} \Sigma^{1/2} \mathbf{Z}_0$, where $\mathbf{Z}_0 \sim \mathbf{N}_n(\mathbf{0}, \mathbf{I})$.

Exercise 3. We have seen in the class that if $Z_1 \sim \mathbf{N}(\mu_1, 1), \dots, Z_p \sim \mathbf{N}(\mu_p, 1)$ are independent random variables, then the distribution of $S = \sum_{i=1}^p Z_i^2$ is called a non-central χ^2 distribution with p degrees of freedom (df) and non-centrality parameter (ncp) $\delta = \sum_{i=1}^p \mu_i^2$. We denote this distribution by $\chi_p^2(\delta)$.

(a) Show that for $Z \sim \mathcal{N}(\mu, 1)$, $\mathbb{E}(Z^2) = 1 + \mu^2$ and $\text{Var}(Z^2) = 2 + 4\mu^2$.

Hint: Z can be written as $Z = \mu + W$, where $W \sim \mathcal{N}(0, 1)$. Use this to find $\mathbb{E}(Z^2)$ and $\mathbb{E}(Z^4)$.

(b) Use part (a) to show that for $S \sim \chi_p^2(\delta)$, $\mathbb{E}(S) = p + \delta$ and $\text{Var}(S) = 2p + 4\delta$. Compare these with the mean and variance of a central χ^2 distribution.

(c) Let $\mathbf{Z} \sim \mathcal{N}_p(\boldsymbol{\mu}, \mathbf{I})$. Show that for any $\mathbf{A} \succeq \mathbf{O}$, $\mathbf{Z}^\top \mathbf{A} \mathbf{Z}$ can be written as a weighted sum of independent non-central χ_1^2 random variables. Also show that $\mathbb{E}(\mathbf{Z}^\top \mathbf{A} \mathbf{Z}) = \text{trace}(\mathbf{A}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ and $\text{Var}(\mathbf{Z}^\top \mathbf{A} \mathbf{Z}) = 2 \text{trace}(\mathbf{A}^2) + 4 \boldsymbol{\mu}^\top \mathbf{A}^2 \boldsymbol{\mu}$.

Hint: Use steps similar to the ones in Exercise 3(a).

We now generalize to more general matrices.

(d) Let $\mathbf{Z} \sim \mathcal{N}_p(\boldsymbol{\mu}, \Sigma)$ where $\Sigma \succ \mathbf{O}$. Show that for any $\mathbf{A} \succeq \mathbf{O}$, if $\mathbf{A} \Sigma \mathbf{A} = \mathbf{A}$, then $\mathbf{Z}^\top \mathbf{A} \mathbf{Z} \sim \chi_r^2(\delta)$, where $r = \text{trace}(\mathbf{A} \Sigma)$ and $\delta = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$.

(e) Use part (d) to verify that $\mathbf{Z}^\top \Sigma^{-1} \mathbf{Z} \sim \chi_p^2(\delta)$, where $\delta = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}$.

(f) (**Hard**) Let $\mathbf{Z} \sim \mathcal{N}_p(\boldsymbol{\mu}, \Sigma)$, where $\Sigma \succeq \mathbf{O}$. Show that $\mathbb{E}(\mathbf{Z}^\top \mathbf{A} \mathbf{Z}) = \text{trace}(\mathbf{A} \Sigma) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ and $\text{Var}(\mathbf{Z}^\top \mathbf{A} \mathbf{Z}) = 2 \text{trace}((\mathbf{A} \Sigma)^2) + 4 \boldsymbol{\mu}^\top \mathbf{A} \Sigma \mathbf{A} \boldsymbol{\mu}$.

Hint: $\mathbf{Z} \sim \mathcal{N}_p(\boldsymbol{\mu}, \Sigma) \stackrel{\mathcal{D}}{=} \Sigma^{1/2} \mathbf{Z}_0 + \boldsymbol{\mu}$, where $\mathbf{Z}_0 \sim \mathcal{N}_p(\mathbf{0}, \mathbf{I})$. What is $\text{Cov}(\mathbf{Z}_0^\top \mathbf{B} \mathbf{Z}_0, \mathbf{c}^\top \mathbf{Z}_0)$?

Exercise 4. Consider the linear model: $\mathbf{y} = \mathbf{X} \boldsymbol{\theta} + \boldsymbol{\epsilon}$. We have seen that under the assumption $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$:

- $\hat{\boldsymbol{\theta}}_{\text{LSE}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \sim \mathcal{N}_p(\boldsymbol{\theta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$.
- $\frac{\text{SSE}}{\sigma^2} \sim \chi_{n-d}^2$.
- SSE and $\hat{\boldsymbol{\theta}}_{\text{LSE}}$ are independent.

The above can be used to find a confidence interval for any linear parametric form $\mathbf{a}^\top \boldsymbol{\theta}$.

- Construct a $100 \times (1 - \alpha)\%$ confidence interval for $\mathbf{a}^\top \boldsymbol{\theta}$.
- Now suppose that we want to predict the value of y at a new data point $\mathbf{x}_{\text{new}}$. How can we do this prediction?
- Instead of a point prediction, can we give a confidence interval for the value of y at $\mathbf{x}_{\text{new}}$? How? Is this conforming to the model?

Exercise 5 (Practical 1, **Graded**). The `windmill` dataset contains measurements of electricity output of wind turbine over 25 separate fifteen minute periods. We are interested in the relation between `direct output` and the `average wind speed` (velocity, measured in miles per hour) during the recording.

- Fit a linear model with `velocity` as covariate and plot the standardized residuals against the fitted values. Do you notice any residual structure missed by the model mean specification? Try fitting a model using the reciprocal of wind speed as covariate. Comment on the adequacy of the models.

- (b) Using both the models in part (a), predict the output of electricity given that the average wind speed in a given period is 5 miles per hour. Provide prediction intervals for your estimates. You may use the R function `predict` to accomplish both the tasks.
- (c) The electricity production should be zero if there is no wind, yet this is not captured by the model linking `output` to `velocity`. Update your model to remove the intercept. What are the consequences of the latter?